Areeb Ahmad Chaudhry

Islamabad, Pakistan

Email | Website | Google Scholar | GitHub | LinkedIn | Hugging Face | Medium

Research Interests

Multimodal learning and AI for health: trustworthy vision-language models that ground their outputs in evidence and know when they are likely wrong; efficient adaptation of foundation models where data and compute are scarce; reliable LLM agents and their evaluation.

Education

National University of Sciences and Technology (NUST), SEECS Nov 2021 – Jun 2025
Bachelor of Software Engineering *Islamabad, Pakistan*

- **Final Year Project (Grade: A):** Radixpert, an anatomy-guided vision-language model for radiology report generation, carried out in the MachVIS Lab. Graded A in both semesters; published at FIT 2025 (IEEE).
- **Relevant coursework:** Machine Learning; Large Language Models; Probability & Statistics; Linear Algebra & ODEs; Discrete Mathematics; Calculus & Analytical Geometry; Complex Variables & Transforms; Numerical Methods; Data Structures & Algorithms; Design & Analysis of Algorithms; Database Systems; Operating Systems; Cloud Computing; Software Quality Engineering.

Higher Secondary School Certificate (Grade 12), Pre-Engineering 2019 – 2021
GCU Lahore | Marks: 1075/1100 (97.7%) *Lahore, Pakistan*

Secondary School Certificate (Grade 10) 2017 – 2019
Garrison Cadet College Kohat | Marks: 1070/1100 (97.3%) *Kohat, Pakistan*

Publications

[1] M. A. R. Farooqi, **Areeb Ahmad Chaudhry**, M. Y. Ghaffar, N. Perwaiz, and H. M. Kiani. "Radixpert: A Multimodal Approach to Medical Imaging." *IEEE Intl. Conf. on Frontiers of Information Technology (FIT)*, 2025. [IEEE] [PDF] [Code]

[2] M. Javed, **Areeb Ahmad Chaudhry**, N. Perwaiz, U. Athar, A. Salahudeen, and M. M. Fraz. "FeLoRA16-SPP: Parameter-Efficient Fine-Tuning of 3D Multimodal LLMs for Radiology Report Generation and Visual Question Answering." *MICCAI 2025 FLARE Challenge (Task 5)*, 2025. [OpenReview]

Technical writing (Medium): LLM as Operating Systems: Agent Memory (2025); Training WAN2.2 for Character and Custom-Style Images (2025); Deploying ComfyUI Workflows on Runpod Serverless (2025).

Research Experience

MachVIS Lab, SEECS, NUST 2024 – Sep 2025
Research Assistant (Jun – Sep 2025); undergraduate researcher during final-year project. Advisor: Dr. Nazia Perwaiz *Islamabad, Pakistan*

FeLoRA16-SPP: 3D multimodal LLM for CT report generation and VQA Jun – Sep 2025

- Adapted a 3D multimodal LLM (M3D-LaMed) to write reports and answer clinical questions about CT scans, training only about 8% of its 4.07B parameters by combining LoRA with a spatial pooling projector.
- Reached an average GREEN score of 0.431 across 18 organ systems, up to 12% above PHI3 and 4% above Med3DVLM, with top scores on 9 organs; global VQA accuracy remained below the baselines.

Radixpert: anatomy-guided VLM for radiology (Final Year Project, Grade A) Sep 2024 – Jun 2025

- Co-developed a model that conditions a language model (MedGemma-4B) on both chest X-ray features and anatomical segmentation masks, so that generated reports are tied to anatomy; trained about 1.8% of parameters with LoRA on 16,000 PadChest studies.
- Reached RadCliQ 0.804, comparable to the strongest baseline (0.810); ablations showed each component's contribution (0.682 $\rightarrow$ 0.755 $\rightarrow$ 0.804), and anatomical grounding reduced left/right errors.

Industry Experience

GoSaaS, AI Engineer | *Remote* Jun 2025 – Present

- Automated overpayment settlement for the finance team of a NYSE-listed US consumer-products company: a 5-page manual workflow became a single request, saving over 10 minutes per settlement.
- Replaced manual lookup for an enterprise client with image-based search over 20,000 CAD drawings, comparing open-source image encoders and embedding models to match part photos to the right drawing; built proof-of-concept data extraction from scanned invoices and industrial diagrams.
- Shipped 8+ production agents with built-in evaluation on the Oracle Marketplace, now used by Oracle partners as templates for their ERP systems; prototyped plain-English (NL2SQL) querying of enterprise databases.

Heartstamp, Founding AI Engineer | *Las Vegas, USA (Remote)* Sep 2025 – Jul 2026

- Designed the product's core multi-agent LLM system (support, design and creative agents), with persistent memory and provider-agnostic routing that let each task switch models without code changes, cutting generation cost.
- Built an evaluation pipeline that benchmarked LLMs (GPT, Gemini, Grok and open-source models such as Kimi) on quality, speed and cost to choose production models, and caught regressions before each release.

Codeacme, AI Engineer (Contract) | *Islamabad, Pakistan (Remote)* Mar 2025 – Sep 2025

- Compared open-source image-generation models (FLUX.1, Stable Diffusion, HunyuanImage, Qwen-Image) and trained 6–7 custom character and style LoRAs for subject-driven generation, including artist-style images for Unsplash.
- Shipped 15+ production ComfyUI workflows (image-to-image, style transfer, inpainting and outpainting, relighting, upscaling, colourisation), deployed to RunPod serverless through a CI/CD pipeline I built, removing always-on compute.

Pam AI, AI Support Engineer (Contract) | *Virginia, USA (Remote)* Dec 2024 – May 2025

- Maintained an 8,000-line production prompt for a voice AI assistant used by hundreds of car dealerships; traced escalated failures in production AWS logs and fixed them by revising prompts and evaluation scripts, reducing hallucinated answers and human escalations.

Tranzbox, Data Scientist | *Melbourne, Australia (Remote)* Jun 2023 – Sep 2024

- Benchmarked classifiers (SVM, random forest, XGBoost, the strongest) and trained A3C reinforcement-learning agents for market prediction, alongside statistical curve fitting and Fourier analysis to find buy and sell points.
- Took models to production: Airflow ETL pipelines on AWS MWAA from data extraction to training and deployment, and GitHub Actions CI/CD releasing models to S3 and AWS Lambda; led a team of 4 engineers.

Selected Projects

- **NUST BankLlama** [GitHub]: a self-hosted banking assistant (LLaMA 3.2-3B fine-tuned with QLoRA) with hybrid retrieval and guardrails against leaking personal data and jailbreaks.
- **Scanned-document question answering** [GitHub]: OCR of scanned tax documents plus retrieval with fine-tuned embedding models, improving retrieval by 4.3%.
- **Research-paper assistant** [GitHub]: retrieval over arXiv papers with semantic chunking, re-ranking and systematic evaluation.

Awards & Leadership

- **3rd of 40**, Devathon speed-programming hackathon, Devsinc Nov 2024
- **Hack Club, NUST Chapter**: Executive, Deputy Director Finance, then Treasurer; led a 24-person team across logistics, finance, decor and liaison, and co-organised a 50-participant student hackathon 2022 – 2024
- **NUST Excursion Society**: Deputy Director Liaison, coordinating trips with the NUST administration 2023
- **SEECS Endowment Fund Society**: Deputy Director, raising funds for deserving students 2022

Technical Skills

Machine learning:	LoRA/QLoRA, vision-language and 3D multimodal models, diffusion models, OpenCV
LLM systems:	LangGraph, LlamaIndex, retrieval-augmented generation, evaluation (RAGAS, DeepEval)
Engineering:	Python, FastAPI, Docker, AWS, PostgreSQL/pgvector, Airflow, CI/CD